

Information Technologies for Industrial Engineers

Backend Application

What is backend?

- The backend is the part of a web application that runs on the server.
- It handles the business logic, database interactions, and server-side processing.
- It is responsible for managing data, processing requests from the frontend, and ensuring that the application functions correctly.

What about `Vue.js` + `Supabase` app?

- This app is built using `Vue.js` for the frontend and `Supabase` for the backend services.
- While `Supabase` is convenient, it may not offer the flexibility and control that a custom backend application can provide such as
 - Custom business logic
 - Complex data processing
 - Integration with other services.

Popularity

- Github stars
- State of JS 2025
- Stack Overflow Developer Survey 2025

Getting started with `express.js`

- Create a new folder for the project.
- Open `VSCode` in that folder (drag folder in VSCode)
- Open a terminal in VSCode.
- `pnpm init`
- Express.js
 - `pnpm i express`

Getting started with `express.js`

- TypeScript
 - `pnpm i typescript tsx`
 - `pnpm approve-builds`
 - `pnpm i -D @types/express @types/node nodemon`
- Create `tsconfig.json` ([Code](#))
- Create `nodemon.json` ([Code](#))
- Update `package.json` ([Code](#))

Main code

- `src/index_1.ts`

API Software

- Postman ([Download](#))
 - Windows: `choco install postman`
 - Mac: `brew install --cask postman`

JavaScript concepts in the code

- Template literals ([Details](#))
- Logical OR (`||`) ([Details](#))
- Nullish Coalescing (`??`) ([Details](#))

HTTP Methods

Method	Purpose
GET	Retrieve data
POST	Create a new resource
PUT	Replace an existing resource
PATCH	Partially update a resource
DELETE	Delete a resource

GET and POST are the most commonly used methods in web applications.

GET

- Used to retrieve data from the server.
- Example: Fetching a list of users from a database.
- Can have query parameters to filter or sort the data.

POST

- Used to create a new resource on the server.
- Example: Submitting a form to create a new user account.
- The request body contains the data for the new resource.

Three ways to send a data in a request

- Query parameters (in the URL)
- Request body (in the body of the request)
- Path Parameters (in the URL path)

JSON

- JSON (JavaScript Object Notation) is a lightweight data interchange format.
- It is easy for humans to read and write, and easy for machines to parse and generate
- Example:

```
{  
  "name": "John Doe",  
  "age": 30,  
  "email": "john.doe@example.com"  
}
```

Common HTTP status codes

Code	Meaning	Typical use
200	OK	Request succeeded
201	Created	New resource was created
204	No Content	Request succeeded without a response body
400	Bad Request	Request data is invalid
401	Unauthorized	Authentication is required
403	Forbidden	Access is not allowed
404	Not Found	Resource does not exist
500	Internal Server Error	Unexpected server failure

Debugging

- Use `console.log()` to print messages to the console for debugging purposes.
- Use VSCode debugger to set breakpoints and inspect variables during code execution.
 - Create `.vscode/launch.json` ([Code](#))
 - Start debugging by pressing `F5` or clicking the green play button in VSCode.

Part 2: Todo API

Starter

Link

The screenshot shows the GitHub interface for the repository 'express-template' (Public). The repository has 1 branch (main) and 0 tags. The file list includes:

File/Folder	Commit Message	Commit Date
.vscode	Initial commit	
src	Initial commit	
.gitignore	Initial commit	
README.md	Update README and package.json with project description and	
nodemon.json	Initial commit	
package.json	Update README and	
pnpm-lock.yaml	Initial commit	
pnpm-workspace.yaml	Initial commit	last week
tsconfig.json	Initial commit	last week

The 'Code' dropdown menu is open, showing options for cloning or downloading the repository. The 'Download ZIP' option is highlighted with a red box.

Options in the 'Code' dropdown:

- Clone (HTTPS, SSH, GitHub CLI)
- https://github.com/it-for-ie-69/express-templat
- Clone using the web URL.
- Open in GitHub Copilot app
- Open with GitHub Desktop
- Open with Visual Studio
- Download ZIP (highlighted)

Main code

- `src/index_2.ts`

JavaScript concepts in the code

- Array method ([Details](#))
 - `filter`
 - `find`
 - `findIndex`
 - `push`
 - `splice`
- Optional chaining (`?.`) ([Details](#))